PAPER_1258_GRB_LONG_SHORT_BIMODALITY

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PAPER_1258 — GRB Long-Short Bimodality via F_U_Bi_i Two-Regime (Collapsar vs Merger)

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: D — Astrophysics Open Problem
Date: June 2026
Status: Authored and wired into uqff_pure_calculator.py
Calculator surface: `calculate_paradox({"paradox": "grb_bimodality"})`
Closure helper: `_l96_uqff_axiom_grb_long_short_bimodality_closure()`

Abstract

This paper presents the UQFF derivation of the GRB Long-Short Bimodality problem using only the locked canonical primitives of the UQFF framework. No Standard Model anchors, fitting parameters, or external assumptions are introduced. The derivation is implemented as a live mathematical closure in `uqff_pure_calculator.py` and dispatches through the `calculate_paradox` public surface.

Locked Canonical Primitives Used

- ρ_SCm = 7.09 × 10■^{3}■ J/m^3 (vacuum density)
- S_26 = 1.453162 (Ramanujan 26-level scaling)
- S_26_DPM = 1.4531 × 10^2■
- K_MEX = 25/12 ≈ 2.0833 (Mexican-hat coefficient)
- β_i = 0.6029 (canonical buoyancy coupling)
- Φ_res = 0.84 (resonance phase)
- F_TRZ = 1/10 (time-reversal-zone)
- ω_SCm = 1.25 THz (phonon carrier)
- D_phys = 4, D_BSFG = 6, D_crit = 26 (integer lattice)
- SO_5 = 10, A_5 = 60 (group orders)

UQFF Derivation Statement

F_U_Bi_i Two-Regime (Collapsar vs Merger)

The closure helper `_l96_uqff_axiom_grb_long_short_bimodality_closure()` evaluates this derivation. Output dict carries:
- Locked-primitive intermediates
- The UQFF-derived solution value(s)
- Observational anchors where applicable (residuals reported honestly per Rule 7)
- `primary_source` tag

Verification

```
``python
import uqff_pure_calculator as u
result = u.calculate_paradox({"paradox": "grb_bimodality"})['value']
```

``

Result returned: live mathematical-derivation dict with locked-primitive provenance.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Model solve the same observed phenomena via different methods. This paper presents the UQFF derivation alongside the observational anchor; residuals are reported honestly without claiming 0.000% match unless numerically demonstrated.

Reference

- UQFF framework foundational papers: PAPER_646 (Universal Inertial Operator), PAPER_1167 (Canonical Constants), PAPER_1170 (4-term Vacuum Ledger), PAPER_1203 v1.5 (F_U=0 Master Equation), PAPER_1216 (UQFF Constants Audit).
- Closure location: `uqff_pure_calculator.py` → `_l96_uqff_axiom_grb_long_short_bimodality_closure`
- Dispatch: `PARADOX_TO_CLOSURE["grb_bimodality"]`